

Economic History II

Colonial legacies in Central Africa

FGV EPGE

Today

- Two studies of colonial Central Africa by Sara Lowes and Eduardo Montero.
- Rubber concessions in the Congo Free State, 1892 to 1906.
- French sleeping sickness campaigns in Cameroon and Equatorial Africa, 1921 to 1956.
- In both, the outcome measured today is **trust**.

Concessions in the Congo

Lowes and Montero (2021), *Concessions, Violence, and Indirect Rule*

The Congo Free State

- Berlin Conference, 1884 to 1885. The European powers accept Leopold II's claims over the Congo basin.
- The stated purpose: free trade, ending the slave trade, a civilizing mission.
- 1885: the Congo Free State, **personal colony** of the king of Belgium.
- 1891 and 1892: all land and all raw materials on it declared property of the state.



A QUAND LE RÉVEILLON DES PEUPLES ?

"Belgium does not need a colony. Belgians are not drawn towards overseas enterprises: they prefer to spend their energy and capital in countries which have already been explored or on less risky schemes ... Still, you can assure His Majesty of my whole-hearted sympathy for the generous plan he had conceived, as long as the Congo does not make any international difficulties for us."

—Walthère Frère-Orban, Liberal Prime Minister of Belgium, 1878–84.^[25]

Why rubber, and why then

- 1890: the pneumatic tire. Demand for natural rubber rises sharply.
- The upper Congo Basin had immense wild rubber. The state had almost no labor and no capacity.
- So Leopold cut the **domaine privé** into concessions for private companies.
- Abir and Anversoise, both 1892: the two largest, both focused on rubber alone.
- The state took 2 percent of profits. Leopold was a majority stakeholder in both.

Boundaries drawn on rivers

- The interior was uncharted, so the contracts used the only salient features available: **river basins**.
- Abir: the Maringa and Loporì basin. Anversoise: the Mongala basin.
- The borders match no present-day political border, no ethnic boundary, no resource deposit.
- The two concessions lasted 14 years, 1892 to 1906.

How the rubber was collected

- Company agents at each post surveyed the villages and counted the adult men.
- **Quotas** were set from that count: about 4 kg of dried rubber every two weeks per man.
- Villages also had to feed and supply the post.
- Most rubber came from the Landolphia vine, delicate and quickly exhausted.
- In the Baringa area, men spent around 10 of every 14 days in the forest.

Enforcement

- Roughly 25 to 80 armed **sentries** at each post, and another 65 to 100 stationed in the surrounding villages.
- Sentries were recruited from other parts of the Congo, chosen so they would use violence against strangers.
- 1903: one Abir post received 17,600 cartridges. A human hand was required for every bullet spent.
- Missing the quota: imprisonment, forced labor, the chicotte, family held for ransom, death.
- July 1902: 44 chiefs imprisoned in the villages around a single post.

Testimony collected by Roger Casement

“When I was still a child, the sentries shot at the people in my village because of the rubber. My father was murdered: they tied him to a tree and shot and killed him, and when the sentries untied him they gave him to their boys, who ate him. My mother and I were taken prisoner. The sentries cut off my mother’s hands while she was still alive. Two days later, they cut off her head.”

Consul sent by Britain to investigate the accusations (Janssens 1904).

Indirect rule, village by village

- The head sentry in each village was the **kapita**. Today kapita is the Lingala word for chief.
- Each kapita recruited 8 to 10 bodyguards, then went after the men of standing.
- Lineage headmen were made to carry soil and rubbish alongside slaves.
- Sentries took over their disputes, their authority and their wealth.
- Morel (1905): headmen “shamelessly degraded in the eyes of their people, made to fetch and carry for soldiers, cast into chains and flung into prison”.

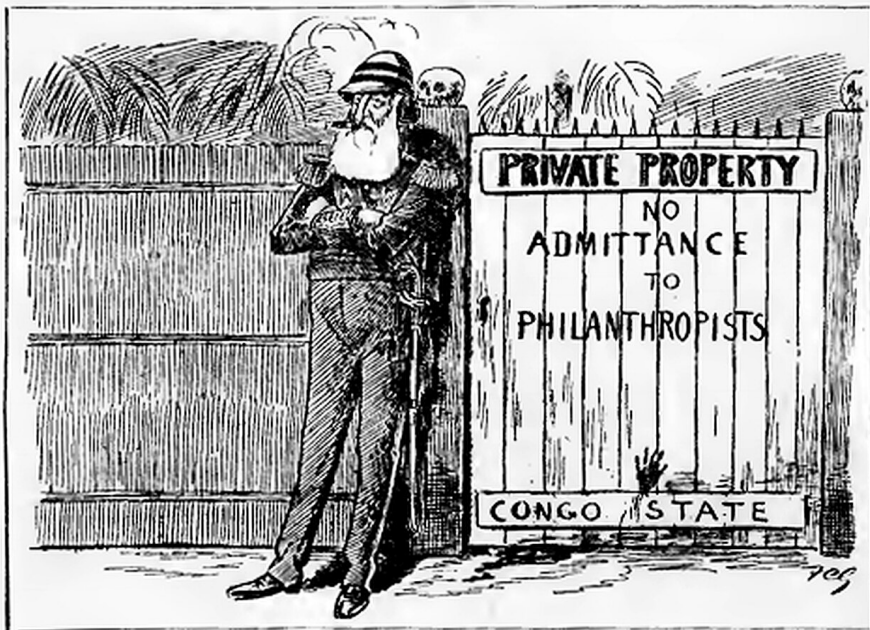
How it ended

- By 1905 the wild rubber of the upper basin was nearly exhausted.
- 1906: Abir and Anversoise leave the Free State, under depleted supplies and European condemnation.
- 1908: the Congo becomes a Belgian colony.
- After 1910: Hevea plantations in Southeast Asia and South America, plus synthetic rubber, collapse the price.
- An estimated **10 million** people died between 1880 and 1920, about half the population.

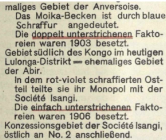


IN THE RUBBER COILS.

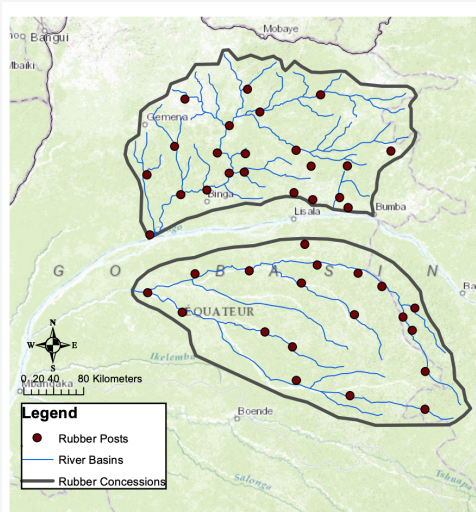
SCENE—The Congo "Free" State.







Abir, Anversoise and the rubber posts



The question

- What did 14 years inside a rubber concession leave behind?
- The concessions were everywhere in colonial Africa, which is exactly why they are hard to study.
- Here the borders were drawn on river basins, before anyone knew what was inside them.
- So compare villages just **inside** the boundary with villages just outside.

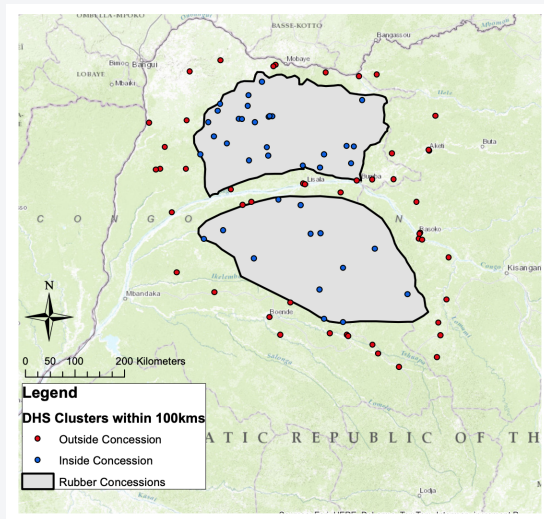
Comparing at a cutoff

- Imagine a scholarship given only to students who score above 70.
- A student with 69 and a student with 71: near identical ability, school, family, motivation.
- One gets the scholarship, the other does not.
- So compare only students **close to the cutoff**.
- A jump in outcomes exactly at 70 is read as the effect of the scholarship.

The same idea on a map

- The cutoff is now a line drawn along river basins.
- Villages a few kilometers apart: same forest, same rivers, same disease environment.
- One side fell under the rubber regime, the other stayed in the free trade zone.
- Villages could not negotiate which side they were on.
- Only villages **near the boundary** enter the comparison.

Villages inside and outside the concessions

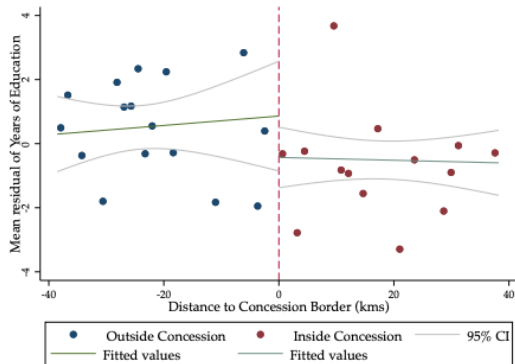


	Within 100 km			Within 50 km			RD Estimates	
	Inside	Outside	SE	Inside	Outside	SE	RD Coefficient	SE
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Geographic Characteristics:								
<i>Elevation</i>	437.26	430.24	(8.10)	440.06	432.46	(5.87)	2.76	(17.39)
<i>Rainfall (Avg.)</i>	72.49	76.42	(4.08)	70.43	74.19	(3.22)	0.01	(7.59)
<i>Rainfall (St. Dev.)</i>	1.31	1.43	(0.07)	1.32	1.41	(0.07)	-0.02	(0.16)
<i>Land Suitability</i>	5.78	7.62	(1.64)	5.01	7.84	(1.94)	-2.26	(2.42)
<i>Ruggedness</i>	4.94	6.25	(1.02)	5.31	5.74	(0.87)	0.26	(2.24)
Obs.	349	504		232	272		894	
River Characteristics:								
<i>Navigable River Density</i>	10.07	10.51	(2.58)	12.20	9.19	(3.41)	1.64	(4.48)
<i>>0 Navigable Rivers</i>	0.21	0.22	(0.05)	0.24	0.19	(0.07)	0.04	(0.08)
<i>>0 Rivers</i>	0.44	0.51	(0.05)	0.38	0.46	(0.07)	-0.11	(0.09)
Obs.	349	504		232	272		894	
Disease Characteristics:								
<i>Malaria Suitability</i>	18.76	18.75	(0.24)	18.71	18.84	(0.19)	-0.23	(0.29)
<i>TseTse Fly Suitability</i>	1.33	1.34	(0.01)	1.32	1.33	(0.01)	0	(0.02)
Obs.	349	504		232	272		894	
Location Characteristics:								
<i>Distance: Kinshasa</i>	767.30	792.81	(37.53)	764.17	776.68	(25.17)	21.82	(58.51)
<i>Distance: Coast</i>	1093.71	1047.54	(19.33)**	1082.01	1064.67	(11.35)*	-18.23	(26.47)
Obs.	349	504		232	272		894	
Pre-Concession Characteristics:								
<i>Num. Enslaved (Atlantic Trade, 1000s)</i>	0.65	2.40	(1.66)	0.91	4.24	(3.22)	-3.92	(6.41)
Obs.	236	314		159	170		573	
Num Ethnic Groups	11	23		10	17		24	
<i>Population Density</i>	1.41	1.19	(0.19)	1.36	1.18	(0.12)	0.09	(0.19)
Obs.	121	187		74	89		329	
Num Ethnic Groups	3	7		3	6		7	
<i>Centralization</i>	0	0.21	(0.10)	0	0.14	(0.07)	-0.04	(0.08)
Obs.	124	280		103	148		426	
Num Ethnic Groups	5	11		5	10		11	
<i>Polygynous</i>	0.46	0.47	(0.11)	0.49	0.45	(0.11)	0.06	(0.21)
Obs.	247	322		173	176		593	
Num Ethnic Groups	7	13		7	12		13	
<i>Hereditary Local Headman Selection</i>	1	0.90	(0.10)	1	0.96	(0.04)	-0.04	(0.03)
Obs.	204	240		142	139		462	
Num Ethnic Groups	5	11		5	10		11	

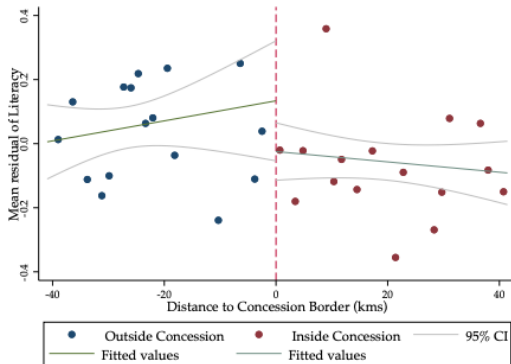
Two sides that look alike

- Elevation, rainfall, ruggedness, land suitability: no difference at the boundary.
- Rivers and navigable rivers: no difference.
- Malaria and tsetse fly suitability: no difference.
- Before the concessions: slaves taken in the Atlantic trade, population density, political centralization, hereditary headmen: no difference.

Inside the concession: less education

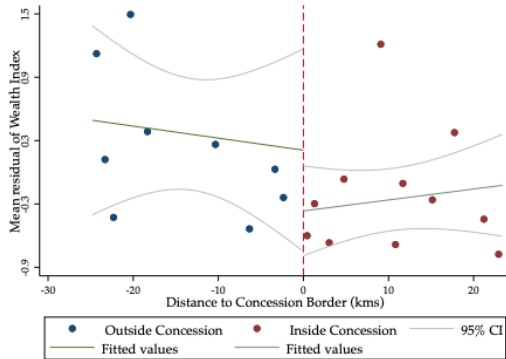


(a) *Years of Education*

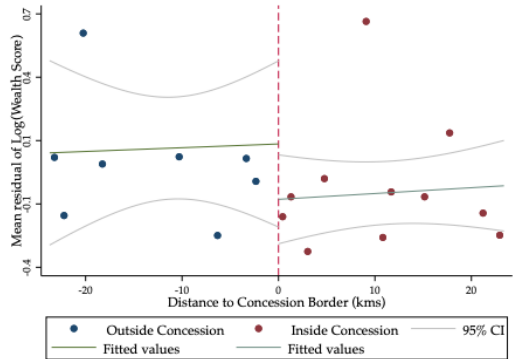


(b) *Literacy*

Inside the concession: less wealth

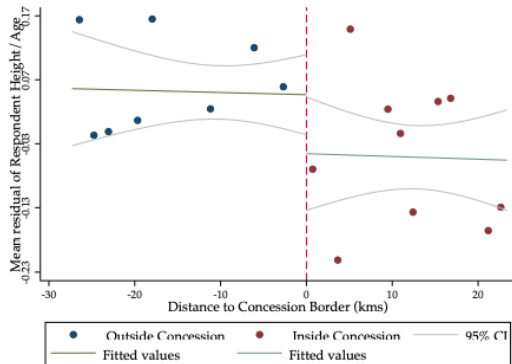


(c) *Wealth Index*

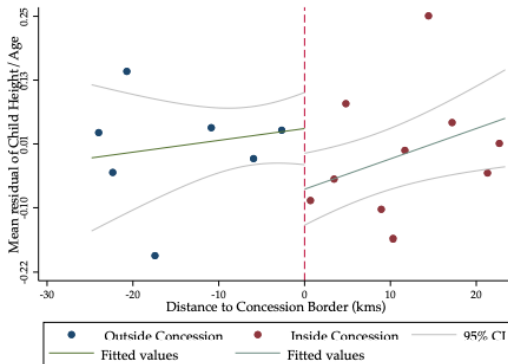


(d) *Log(Wealth Score)*

Inside the concession: worse health



(e) *Respondent Ht./Age*



(f) *Child Ht./Age*

The magnitudes, a century later

- Just inside the old boundary: about **1.4 fewer years of education**.
- About 26 percent less wealth, roughly half a standard deviation.
- Children and women about 9 percentage points lower on height for age.
- Household surveys from 2007 and 2014, more than a century after the concessions were drawn.

Why?

Chiefs, and what people came to believe

The fieldwork in Gemena

- Gemena is a town on the edge of the former Anversoise concession, built after the rubber era.
- Everyone there is a migrant or the descendant of migrants.
- July and August 2015: **520** randomly sampled respondents, two visits per household.
- About half trace their village of origin to inside a former concession, half to just outside.
- Everyone now lives under the same institutions, so what differs is what they brought with them.

Their villages of origin, today

Panel A: Village Development

	<i>Village Public Goods Index</i>		<i>Village Subjective Ratings Index</i>	
	(1)	(2)	(3)	(4)
Inside Concession	−0.127* (0.074)	−0.115** (0.055)	−0.880*** (0.252)	−0.434** (0.176)
Bandwidth Choice	Optimal	Wide	Optimal	Wide
Observations	267	304	147	212
Clusters	136	221	87	162
Bandwidth	34.40	100.00	22.55	100.00
Mean Dep. Var.	−0.003	0.003	0.062	0.068
SD Dep. Var.	0.285	0.291	0.740	0.690

The chiefs

- A village chief is either elected by villagers or inherits the position within a ruling lineage.
- Inside the former concessions, chiefs are **35 percentage points** less likely to have been elected.
- The companies anointed ruling lineages, in some cases handing out medals as proof.
- Those chiefs also do less: road maintenance, arbitration of conflict, upkeep of schools.
- With a weak national government, the chief is who provides public goods or does not.

Fewer elected chiefs, fewer public goods

<i>Panel B: Chief Quality and Accountability</i>				
	<i>Chief Elected</i>		<i>Chief Public Good Index</i>	
	(1)	(2)	(3)	(4)
Inside Concession	-0.423** (0.182)	-0.345*** (0.129)	-0.277** (0.136)	-0.194* (0.106)
Bandwidth Choice	Optimal	Wide	Optimal	Wide
Observations	189	255	250	296
Clusters	104	189	119	210
Bandwidth	26.85	100.00	28.26	100.00
Mean Dep. Var.	0.503	0.506	-0.083	0.016
SD Dep. Var.	0.502	0.501	0.519	0.494

What the survey asked about trust

- How much do you trust: your family, your neighbors, your own tribe, another tribe, another nationality, someone you meet for the first time.
- How close do you feel to: your village of origin, the people of Gemena, your own tribe, your age set.
- Both on scales, both asked of everyone in the sample.

More trust and more closeness inside

	<i>Panel A: Trust and Closeness</i>			
	<i>Trust Index</i>		<i>Closeness Index</i>	
	(1)	(2)	(3)	(4)
Inside Concession	0.299 (0.236)	0.313*** (0.120)	0.521** (0.211)	0.551*** (0.135)
Bandwidth Choice	Optimal	Wide	Optimal	Wide
Observations	304	465	299	465
Clusters	151	313	158	313
Bandwidth	23.61	100.00	24.95	100.00
Mean Dep. Var.	0.142	0.041	0.156	0.045
SD Dep. Var.	0.698	0.707	0.717	0.717

Two games played for money

- Dictator game: a player receives a sum and decides how much to send to an anonymous second player.
- Effort task: player 1 earns money by tapping a moving dot on a tablet for five minutes, at 100 Congolese francs per 10 successful taps, about 700 francs on average.
- Player 2 is then told what player 1 earned, and decides what share of it to take.
- That share measures **respect for earned income**.

Giving is the same, taking is not

<i>Panel C: Experimental Measures of Sharing Norms</i>				
	<i>Dictator Game: Share Sent</i>		<i>Effort Task: Share Redistributed</i>	
	(1)	(2)	(3)	(4)
Inside Concession	-0.018 (0.025)	0.001 (0.020)	0.067** (0.030)	0.053** (0.023)
Bandwidth Choice	Optimal	Wide	Optimal	Wide
Observations	258	438	368	437
Clusters	135	300	193	300
Bandwidth	21.48	100.00	37.12	100.00
Mean Dep. Var.	0.449	0.445	0.410	0.405
SD Dep. Var.	0.122	0.123	0.134	0.134

Reading the two games

- Amount sent in the dictator game: no difference between the two sides.
- Share taken from another player's earned income: about **5 percentage points higher** inside.
- Matches the survey answers: income should be shared, whether it came from work or from luck.
- And matches the data: less income inequality within villages inside the former concessions.

Institutions and culture as substitutes

- Worse chiefs and fewer public goods, but more trust and more sharing.
- As local institutions stopped working, the return to relying on each other went up.
- The two then hold each other in place: chiefs face less accountability because people lean on informal sharing instead.
- Lowes and Montero read this as a **low-development equilibrium**.

Concessions were the normal form

- Concessions to private companies existed in French, British, Belgian, German and Portuguese Africa.
- All of Rhodesia was granted to the British South Africa Company.
- Outside Africa: the East India Company, the Dutch East India Company in Indonesia, the United Fruit Company in much of Central America.
- In Latin America, the closest comparison is the Peruvian **mita**, which we come back to at the end.

Colonial medicine

Lowes and Montero (2021), *The Legacy of Colonial Medicine in Central Africa*

The puzzle

- Sub-Saharan Africa accounts for 90 percent of all malaria deaths and 70 percent of people living with HIV.
- Vaccines and treatment exist, are effective, and are often free.
- Take-up is still low. About 20 million children worldwide are not vaccinated.
- Usual explanations: money, present bias, distance to a clinic.
- This paper adds **mistrust**, and gives it a date.

Sleeping sickness

- Human African trypanosomiasis, transmitted by the bite of the **tsetse fly**, lethal if untreated.
- First stage: joint pain, headaches, fever, swollen lymph nodes.
- Second stage: the nervous system, extreme lethargy, death.
- Central and West Africa have the chronic form, *Trypanosoma brucei gambiense*.
- Epidemics grew as Europeans pushed into rural areas.

The mobile teams

- Cameroon, from 1921. Equatorial Africa, from 1927: the *Service de la prophylaxie de la trypanosomiase*.
- One French military doctor, African nurses, two white corporals, African soldiers, and porters to carry the equipment.
- Villagers were examined **by force**, often at gunpoint: neck palpations, blood tests, spinal taps.
- Deciding which stage of the disease a patient was in was unreliable in the field.
- Cameroon, 1928: about 660,000 people examined in one year, 17 percent recorded as sick.

Atoxyl

- Arsenic based. The name means nontoxic.
- The dose needed to clear the parasite was almost the dose that would kill the patient.
- **Blindness in up to 20 percent** of patients, and it was given regardless of diagnosis.
- It worked in the first stage of the disease. In the second it could hasten death.
- The French kept using it into the 1930s, a decade after other colonies stopped.

Why they kept it

- Cheap: almost **seven times cheaper** than the next cheapest drug.
- Easy for a nurse to administer in the field, and fairly stable in the tropics.
- Tryparsamide, the less toxic alternative, abscessed if the needle or the skin was not perfectly clean.

“Atoxyl’s grip remained because its price and convenience were unbeatable. The health service tolerated accidents and blindness because they did not usually appear at the moment of injection, unlike an abscess from a faulty tryparsamide shot.” (Headrick 1994)

Lomidine

- Given as prevention, to **everyone** in a village, every six months.
- Painful, with dizziness and low blood pressure. Whole villages were made to rest under supervision afterwards.
- Gangrene at the injection site, and deaths. Considered too dangerous to use on Europeans.
- It was eventually shown not to prevent anything.
- A French doctor from the campaigns, in 1974: the injections were “pointless, dangerous, and therefore pointlessly dangerous”.

An Eton song about the campaigns

“The injection against sleeping sickness was too painful

The injection against sleeping sickness was too painful

They gave me an injection in the head

They gave me an injection in the neck

They gave me an injection in the back

...

They ask me to go draw water from the well

If I drag my feet

The policemen hit me on the head

The injection against sleeping sickness was too painful”

The memory is documented

- Cameroon, 1990: resistance to a tetanus campaign, which had “awakened negative collective memories of French colonial efforts to wipe out sleeping sickness”.
- Nola region of the Central African Republic, late 1980s: rumors still circulating that the injections brought death.
- Reused needles: Cameroon today has one of the highest hepatitis C rates in the world, and epidemiologists link it to the campaigns.
- Pépin argues the campaigns may also have helped HIV spread before anyone had identified it.

The hypothesis

- For many villagers, the campaigns were their **first contact** with modern medicine.
- Forced, painful, and largely useless.
- Prediction: lower demand for medicine today, where the teams went most.

Cameroon and former French Equatorial Africa



FIGURE 1. MAP OF CAMEROON AND FORMER FRENCH EQUATORIAL AFRICA

The archives

- Every colony filed an annual health report with France. They sit in the *Service Historique de la Défense*, the military archives.
- January 2014: the authors collected them and digitized hundreds of tables.
- Cameroon 1921 to 1950, by ethnicity and district. Equatorial Africa 1927 to 1956, by subdistrict.
- Each table: people in the area, people visited, new and old cases, lumbar punctures, injections by drug.

Figure B1: Examples of Reports

Table 1. The number of cases of dengue fever and dengue haemorrhagic fever by age group and sex in Bangkok, 1997-2000									
Year	Age group	Sex	Dengue fever		Dengue haemorrhagic fever		Total		Total
			Number	%	Number	%	Number	%	
1997	0-4	M	1,234	100	12	100	1,246	100	1,246
	0-4	F	1,156	100	10	100	1,166	100	1,166
	5-14	M	1,089	100	15	100	1,104	100	1,104
	5-14	F	1,012	100	18	100	1,030	100	1,030
1998	0-4	M	1,345	100	25	100	1,370	100	1,370
	0-4	F	1,289	100	22	100	1,311	100	1,311
	5-14	M	1,156	100	20	100	1,176	100	1,176
	5-14	F	1,089	100	18	100	1,107	100	1,107
1999	0-4	M	1,456	100	30	100	1,486	100	1,486
	0-4	F	1,389	100	28	100	1,417	100	1,417
	5-14	M	1,234	100	25	100	1,259	100	1,259
	5-14	F	1,156	100	22	100	1,178	100	1,178
2000	0-4	M	1,567	100	35	100	1,602	100	1,602
	0-4	F	1,489	100	32	100	1,521	100	1,521
	5-14	M	1,345	100	28	100	1,373	100	1,373
	5-14	F	1,234	100	25	100	1,259	100	1,259

(a) Example of Archival Data
from Gabon (1954)

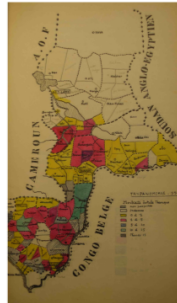
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(b) Example of Archival Data
from Cameroon (1934)

Figure B2: Examples of Maps



(a) Regions Visited in Cameroon (1941)



(b) Sleeping Sickness Prevalence AEF (1944)

Measuring exposure

- For each region: the **share of the 30 years** in which the mobile teams came.
- Visits range from none to more than 20 years.
- The average region was visited 15 years.
- Northern Chad was never visited: no tsetse fly, so no sleeping sickness.

Visits between 1921 and 1956

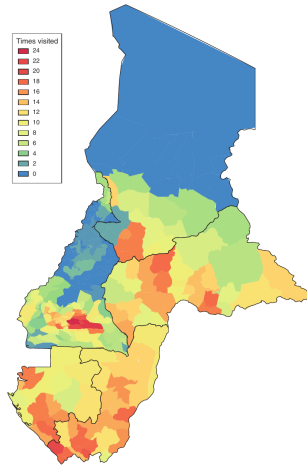


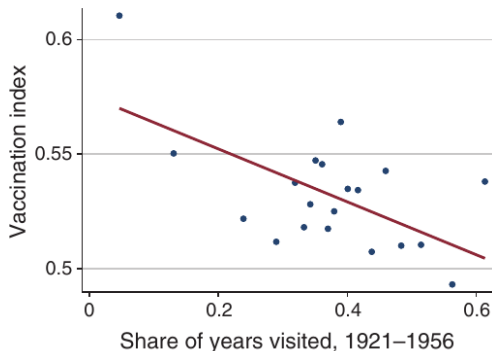
FIGURE 2. SLEEPING SICKNESS VISITS BETWEEN 1921–1956

Two outcomes today

- **Vaccination**: share completed of nine vaccines for children under five. Sample average about 53 percent.
- **Blood test refusal**: whether a respondent refuses a free finger prick for anemia or HIV. About 4.8 percent refuse.
- The test costs nothing and the anemia result comes back in minutes with advice on treatment.
- So refusal is a choice, not a price. The authors read it as revealed mistrust.

Where the teams went most

Panel A. Vaccination index



Panel B. Blood test refusal

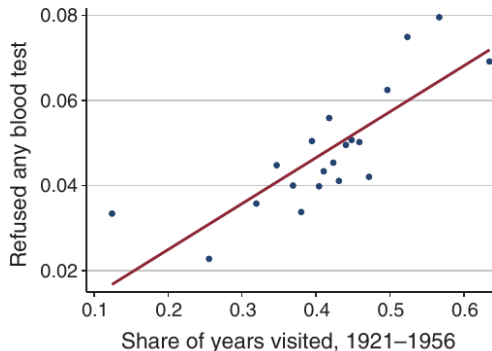


FIGURE 2. DISCOVERY: SHARE OF YEARS VISITED

The correlation

- A region visited 15 years: vaccination **5.8 percentage points** lower.
- The same region: blood test refusals 5.4 percentage points higher, against an average of 4.8 percent.
- Holding fixed temperature, rainfall, land and disease suitability, initial sleeping sickness rates, exposure to the Atlantic slave trade, missions, schooling and wealth.

Why a correlation is not enough

- The teams chose where to go. They may have avoided places that were already resistant.
- Or something else may drive both the visits and health behavior today.
- Tsetse suitability cannot separate these places: almost the whole region is highly suitable.
- So we need something that moved the teams around for a reason unrelated to trust.

Cassava and millet

- Colonial doctors linked cassava to sleeping sickness. Before the tsetse fly was identified in 1903, some thought eating cassava caused it.
- Cassava must be soaked before it is edible, because otherwise it holds cyanide.
- The soaking pits sat on river banks. A 1909 report named those **cassava holes** as a main place people met the tsetse fly.
- Cassava also yields four times the calories per acre of millet, so less land is cleared, leaving more tsetse-harboring bush.

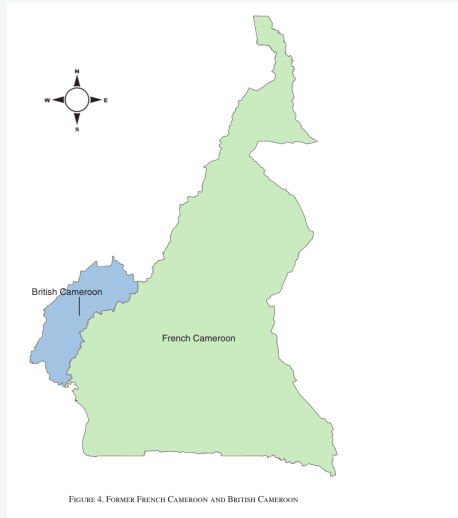
What the soil comparison gives

- Soil suited to cassava rather than millet predicts the sleeping sickness rates the teams recorded, and predicts where they went.
- Comparing places on that basis, a region visited 15 years has vaccination **12.6 percentage points** lower.
- And blood test refusals 8.2 percentage points higher.
- Larger than the plain correlations, consistent with campaigns steering away from places that were already resistant.

But is it the soil itself?

- 1884: Cameroon becomes a German colony.
- 1916: German forces surrender to the British, invading from Nigeria.
- 1919: a League of Nations mandate splits Cameroon between France and Britain.
- The British ran their strip by indirect rule, described as “one of benign neglect”, and ran no medical campaigns.
- 1961: southern British Cameroon votes to join Cameroon, the northern strip joins Nigeria.

French Cameroon and British Cameroon



Same country, same soils, one side treated

TABLE 4—FALSIFICATION TEST: FORMER BRITISH AND FORMER FRENCH CAMEROON

	Instrument falsification test: Suitability for cassava relative to millet							
	DHS vaccination index				Blood test refused			
	Former British Cameroon		Former French Cameroon		Former British Cameroon		Former French Cameroon	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Relative suitability: cassava versus millet	0.0044 (0.0046)	0.0045 (0.0045)	-0.0104 (0.0053)	-0.0111 (0.0047)	-0.0016 (0.0033)	-0.0018 (0.0032)	0.0062 (0.0031)	0.0067 (0.0031)
Geography and climate controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Disease suitability controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Colonial controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Contemporary controls	No	Yes	No	Yes	No	Yes	No	Yes
<i>p</i> -value: Former British versus French coefficients	—	—	0.035	0.017	—	—	0.083	0.057
Bandwidth	50 km	50 km	50 km	50 km	50 km	50 km	50 km	50 km
Observations	2,019	2,019	3,605	3,605	4,867	4,867	7,891	7,891
Clusters	141	141	241	241	141	141	241	241
Mean dependent variable	0.774	0.774	0.743	0.743	0.028	0.028	0.062	0.062

Within 50 km of the old border: cassava soil predicts nothing on the British side, and predicts lower vaccination and more refusals on the French side.

The same test, wider

- Repeat it across British and Portuguese colonies, which ran no equivalent campaigns.
- There the pattern is **8 to 10 times weaker**, and effectively zero.
- Cassava soil predicts vaccination and blood test refusals only where the campaigns actually went.

Access, or demand?

- Afrobarometer for Cameroon and Gabon, rounds 5 and 6.
- No difference in whether a health clinic sits in the respondent's area.
- Among those who tried, no difference in how hard it was to get treatment.
- More exposed places: more people with **no contact** with a clinic in the past year, and more who went without treatment.
- Supply looks the same on both sides. What differs is whether people go.

Less use of the health system today

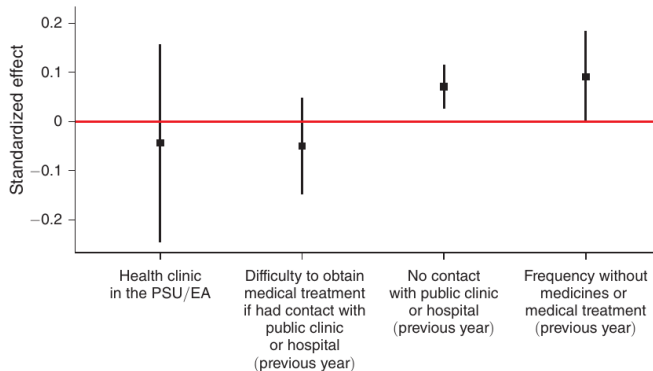


FIGURE 5. OLS ESTIMATES: ACCESS TO HEALTH CENTER AND INTERACTIONS WITH HEALTH SYSTEM

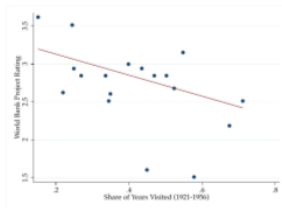
Notes: Data are from the Afrobarometer for Cameroon (Round 5 and Round 6) and Gabon (Round 6). See online Appendix Table B19 for more information on variable definitions and for the results in table format.

Does it matter for policy today?

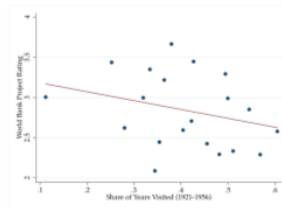
- AidData: World Bank projects approved between 1995 and 2014, each rated from highly unsatisfactory to highly satisfactory.
- In more exposed areas, **health projects** are rated worse. Projects in other sectors are not.
- The size: a health project in an average-exposure area moves from moderately satisfactory to moderately unsatisfactory.
- Exposure does not predict whether an area gets a project at all, or whether a project gets rated.

Health, not projects in general

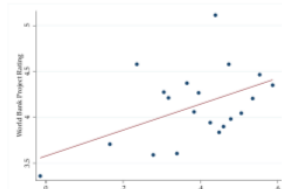
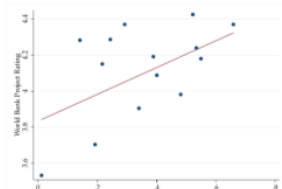
Figure B15: Times Prospected and World Bank Project Outcomes for Health and Non-health Projects



(a) Health Projects, No controls



(b) Health Projects, Full controls



Mistrust of medicine, or of everything?

- Afrobarometer also asks about trust in neighbors, parents, people you know, most people.
- And in the president, parliament, the electoral commission, the police, the army, the courts, traditional and religious leaders.
- Exposure to the campaigns predicts **none** of them.
- The mistrust is confined to the medical domain.

How it travels between generations

- The Cameroon data are by ethnic group, and the 2004 survey records each respondent's ethnicity.
- Your **own** ethnic group's exposure: 8.79 percentage points higher chance of refusing the blood test.
- The exposure of the other ethnic groups around you: positive too, but about half the size.
- Consistent with transmission from parents, without ruling out transmission from neighbors.

A Brazilian comparison

- Rio de Janeiro, 1904: compulsory smallpox vaccination, with health teams entering homes.
- The *Revolta da Vacina* followed, days of rioting in November.
- Same ingredients: a public health measure, imposed without consent, in a city with no reason to trust the state.
- For discussion: does anything of that survive in vaccination behavior today?

Two legacies of colonial coercion

- **Rubber concessions:** less education, less wealth, worse health, chiefs who are not elected, and more trust among neighbors.
- **Medical campaigns:** less vaccination, more refusal of a free blood test, worse health projects, and no effect on any other kind of trust.
- One intervention ended in 1906, the other in 1956. Both are still measurable in survey data.
- In the Congo, institutions and culture moved in opposite directions. In Equatorial Africa, the damage stayed inside one domain.

Other boundaries

Two more comparisons across a line on the map

THE PERSISTENT EFFECTS OF PERU'S MINING *MITA*

BY MELISSA DELL¹

This study utilizes regression discontinuity to examine the long-run impacts of the *mita*, an extensive forced mining labor system in effect in Peru and Bolivia between 1573 and 1812. Results indicate that a *mita* effect lowers household consumption by around 25% and increases the prevalence of stunted growth in children by around 6 percentage points in subjected districts today. Using data from the Spanish Empire and Peruvian Republic to trace channels of institutional persistence, I show that the *mita*'s influence has persisted through its impacts on land tenure and public goods provision. *Mita* districts historically had fewer large landowners and lower educational attainment. Today, they are less integrated into road networks and their residents are substantially more likely to be subsistence farmers.

KEYWORDS: Forced labor, land tenure, public goods

The mita in Peru

mercury required to refine silver ore. Beginning in 1573, indigenous villages located within a contiguous region were required to provide one-seventh of their adult male population as rotating *mita* laborers to Potosí or Huancavelica, and the region subjected remained constant from 1578 onward.⁴ The *mita* assigned 14,181 conscripts from southern Peru and Bolivia to Potosí and 3280 conscripts from central and southern Peru to Huancavelica (Bakewell (1984, p. 83)).⁵ Using population estimates from the early 17th century (Cook (1981)), I calculate that around 3% of adult males living within the current boundaries of Peru were conscripted to the *mita* at a given point in time. The percentage of males who at some point participated was considerably higher, as men in subjected districts were supposed to serve once every 7 years.⁶

The mita in Peru

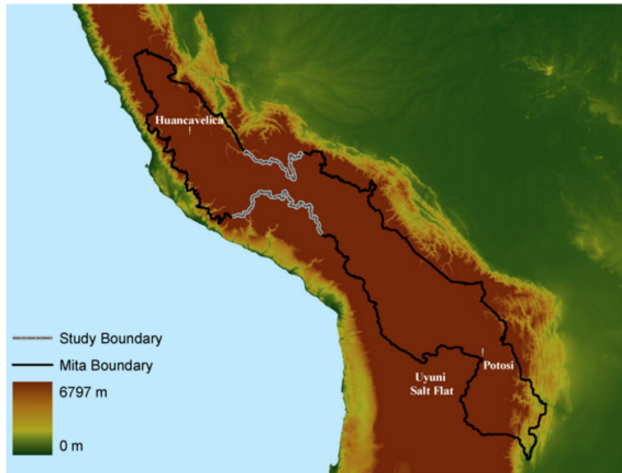
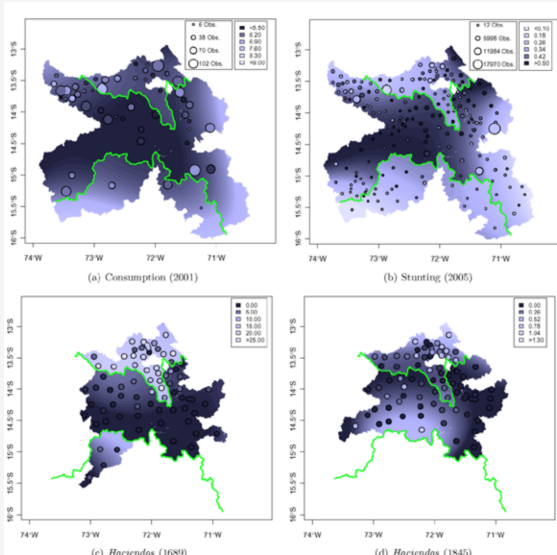
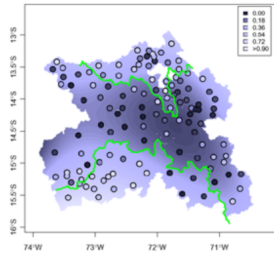


FIGURE 1.—The *mita* boundary is in black and the study boundary in light gray. Districts falling inside the contiguous area formed by the *mita* boundary contributed to the *mita*. Elevation is shown in the background.

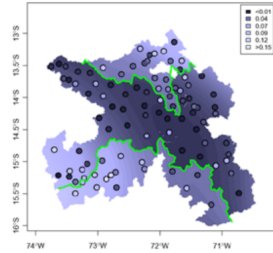
The mita in Peru



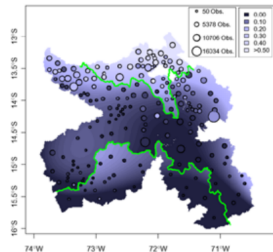
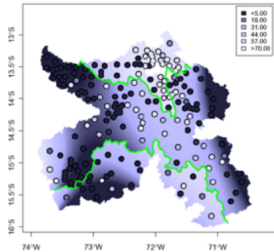
The mita in Peru



(e) Haciendas (1940)



(f) Education (1876)



The mita in Peru

Check for updates

The long shadow of history? the impact of colonial labor institutions on economic development in Peru

Leticia Abad¹ · Noel Maurer²

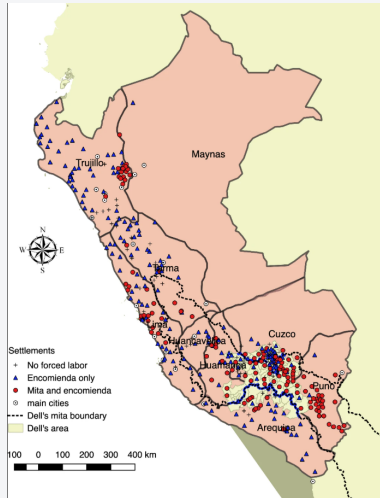
Accepted: 26 November 2024 / Published online: 16 January 2025

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Abstract

We examine a canonical case of forced labor: the *mita* and the *encomienda* in colonial Peru. The *mita* was a labor draft designed to provide workers for mines, churches, and public works in colonial Peru and Bolivia. The *encomienda* granted a select group of Spanish settlers the right to extract labor and tribute from indigenous peoples. We examine the impact of forced labor using a dataset of 500 indigenous settlements scattered across modern-day Peru. We find that forced labor gravely impacted the Peruvian communities subjected to it, but the effects nearly dissipated before the end of the colonial period (1532–1811). We test for a possible “reversal of fortune” in the postcolonial period by looking at an array of variables for the 19th to the 21st centuries (literacy, access to land, road density, and luminosity) and find no significant differences. The results hold when we examine the *mita* and *encomienda* separately. The mechanisms that caused its impact to fade were migration, growing outside options for indigenous labor, and opposition from the Crown and new Spanish settlers who lacked access to forced labor.

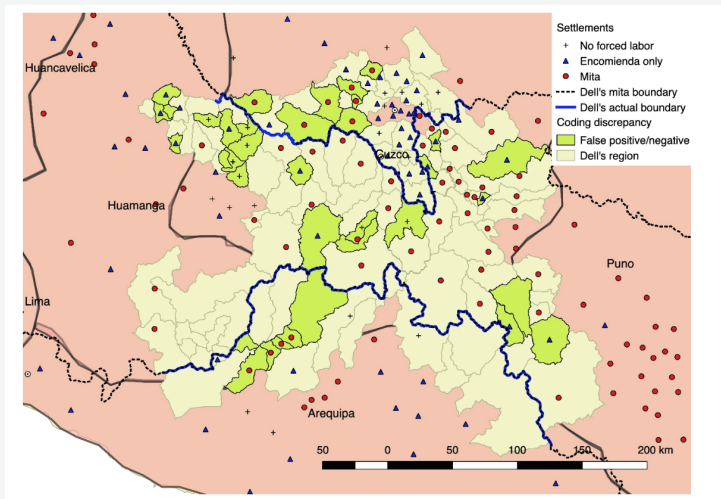
The mita in Peru



The mita in Peru

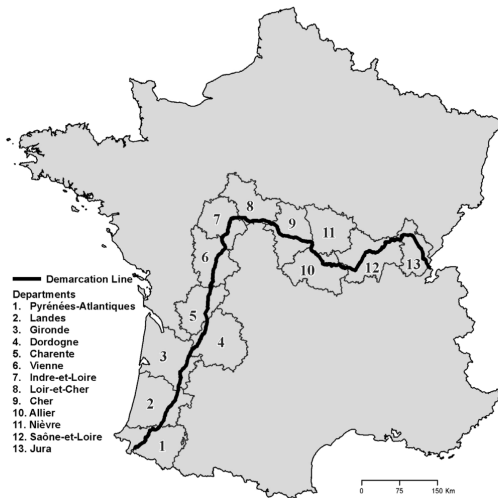
Readers may wonder why our results differ from the findings in Dell (2010) and Carpio and Guerrero (2021).⁴ First, our analysis takes into account not just the mining *mita* but other forms of labor coercion. Second, if we look just at the *mita*, we find that there was no discrete mining *mita* boundary or contiguous mining *mita* area.⁵ Our identification shows there was a patchwork of subject and exempt communities. In addition, our study area comprises the whole of modern Peru, while Dell (2010) examined the Potosí and Huancavelica mining *mitas* in a limited portion of southern Peru (see Sect. 3). In appen-

The mita in Peru



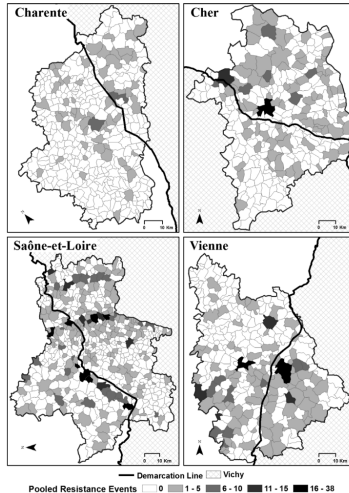
France: the demarcation line

FIGURE 1. Map of the Demarcation Line across Intersected French Departments



France: the demarcation line

FIGURE 2. Resistance Events across Departments



The Hatched Portion of Each Map Represents the Vichy Zone.

France: the demarcation line

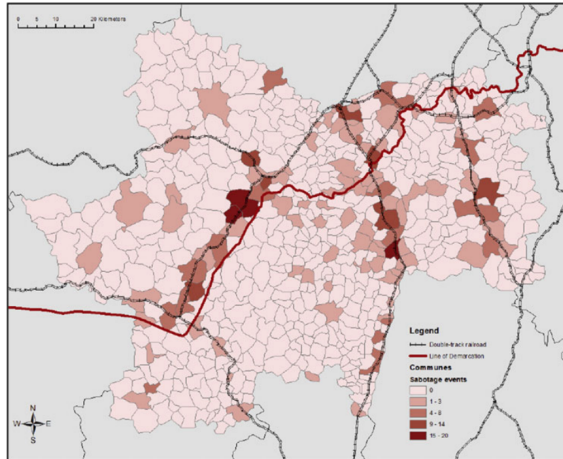
Map 1

The Line of Demarcation (LoD) and Nantes-Tours-Belfort and Paris-Tours-Bordeaux railroads



France: the demarcation line

Map 4
Double-track railways and sabotage in Saône-et-Loire, 1940–1944



France: the demarcation line

Map 2
Double-track railways and sabotage in Charente and Vienne, 1940–1944

